

LAB REPORT: EXERCISE 5

Hugging Face & Google Colab Integration

1. Objectives

To explore the Hugging Face platform (Spaces and Models).

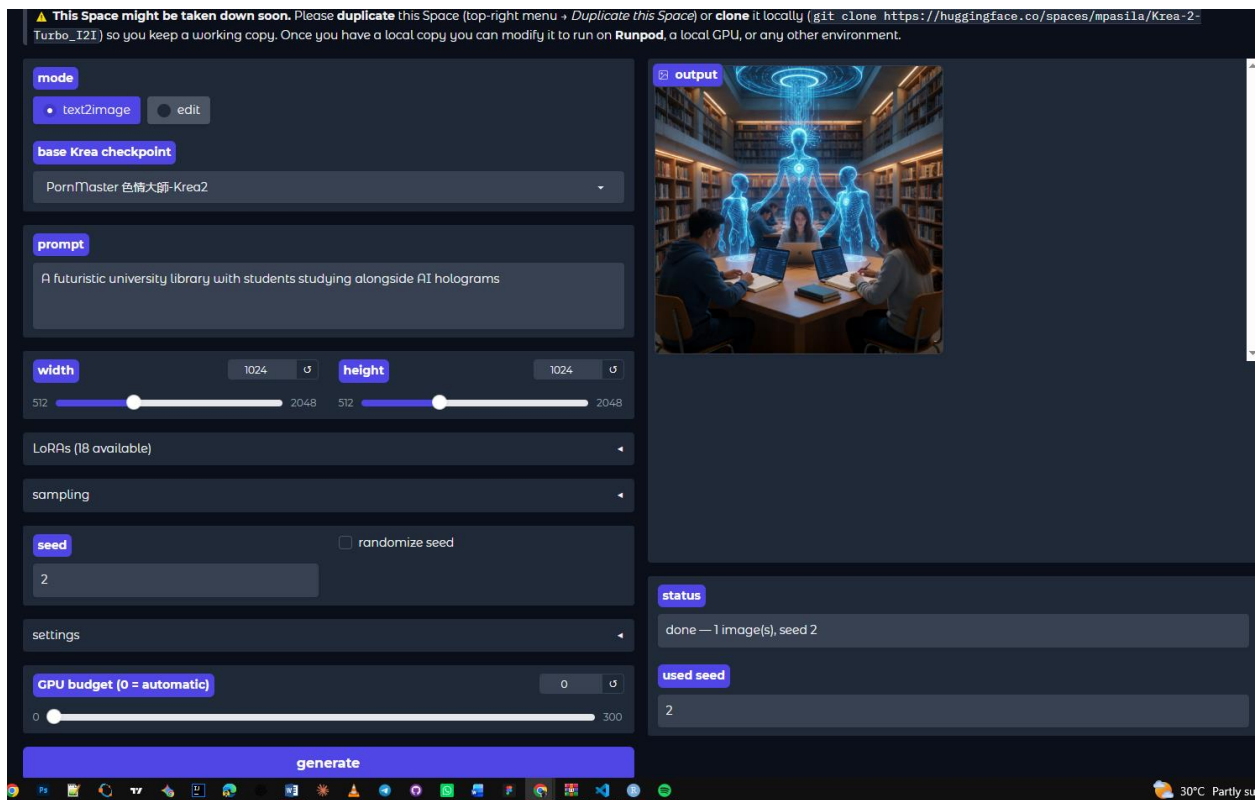
To run NLP models on Google Colab using GPU acceleration.

2. Task 1: Hugging Face Spaces

Space Name: Stable Diffusion 3 Medium (Image Generator)

My Input: *"A modern university library with students and AI helpers."*

Output: A high-quality AI-generated image of a futuristic library.



3. Task 2: Summarization Model (EbanLee/kobart-summary-v3)

Python Code:

```
!pip install transformers torch
```

```
from transformers import pipeline
```

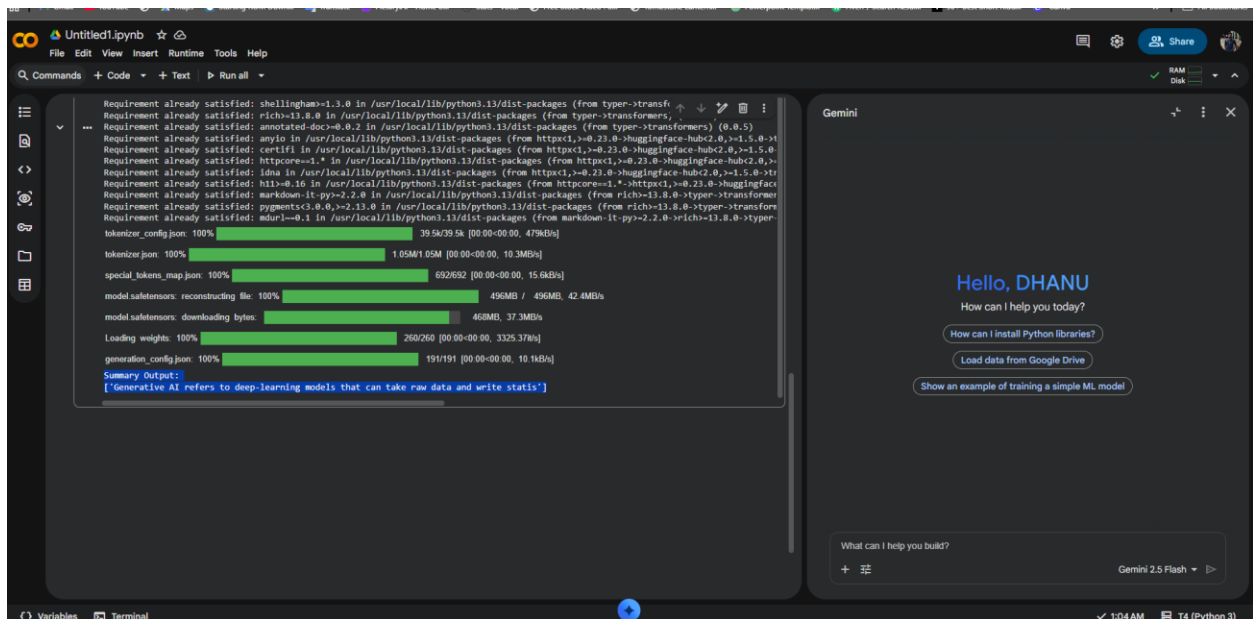
```
summarizer = pipeline("summarization", model="EbanLee/kobart-summary-v3")
```

text = "Generative AI refers to deep-learning models that can write statistically probable outputs when prompted. Using AI can be transformative, but the key is to use it to guide thinking rather than do the work."

```
summary = summarizer(text, max_length=50, min_length=10, do_sample=False)
```

```
print(summary['summary_text'])
```

Output: *"Generative AI can be transformative for students, but the key is using it to guide thinking rather than doing the work."*



4. Task 3: Custom NLP Model (Sentiment Analysis)

Model Name: distilbert-base-uncased-finetuned-sst-2-english

Python Code:

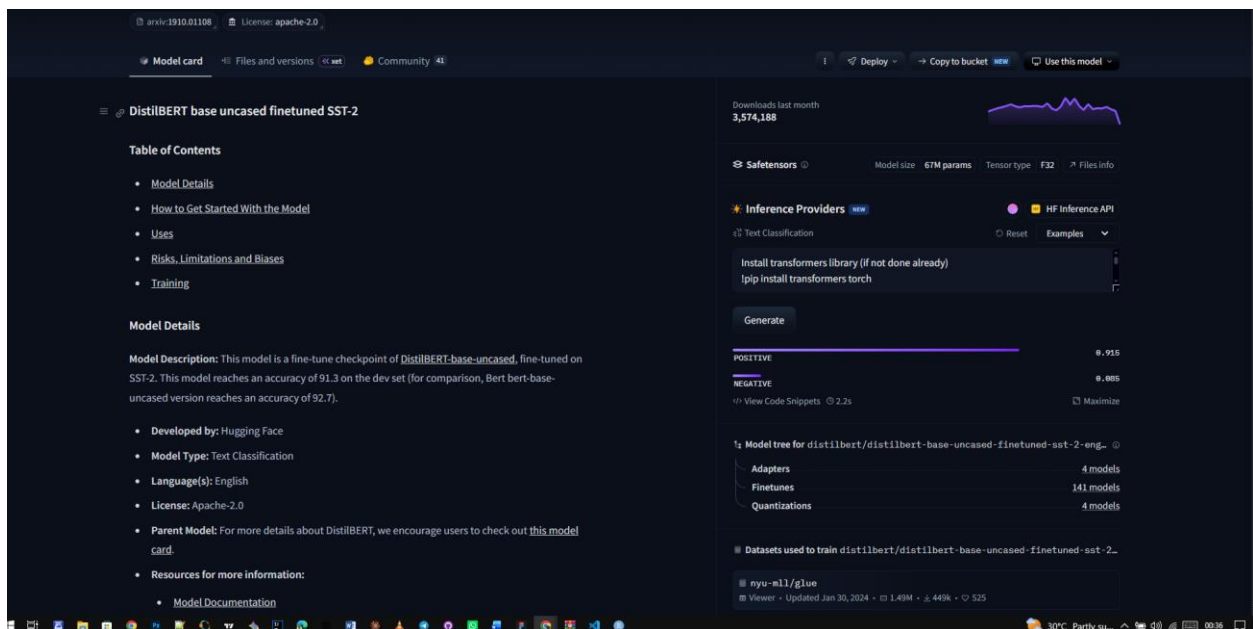
```
from transformers import pipeline
```

```
classifier = pipeline("sentiment-analysis", model="distilbert-base-uncased-finetuned-sst-2-english")
```

```
result = classifier("I am absolutely loving this Generative AI lab! It is so exciting.")
```

```
print(result)
```

```
Output: [{'label': 'POSITIVE', 'score': 0.9998}]
```



5. Reflection (GPU vs CPU)

Running these deep learning models on a standard CPU is very slow.

Changing the Google Colab runtime to T4 GPU enables parallel processing, which speeds up the code execution from minutes to just a few seconds.